

BANNARI AMMAN INSTITUTE OF TECHNOLOGY*Department of Electronics and Communication Engineering (ECE)*

CHRONOBIT '26

Digital Clock Design Hackathon

PROJECT DOCUMENTATION

Digital Clock Using CD4026 Decade Counter/Driver ICs and NE555 Timer

Team Name

Quantum Spark

Event

ChronoBIT '26 — Digital Clock

1. Abstract

This document presents the design, construction, and working of a self-contained digital clock circuit built for Chronobit '26, entirely from combinational and sequential logic ICs, without the use of a microcontroller. The circuit generates its own one-second timebase using an NE555 timer configured as an astable multivibrator, and drives six cascaded CD4026 decade counter / seven-segment driver ICs to display a running six-digit count on six common-cathode seven-segment displays. A single push-button switch provides a manual reset, clearing every digit back to zero. Unlike designs that pair a separate BCD counter with a separate 7-segment decoder, the CD4026 integrates counting and display-driving into one chip, which keeps the component count and wiring density noticeably lower while still demonstrating the same core sequential-logic concepts.

2. Objective

- To design and build a working digital counter/clock display using only standard CMOS logic ICs (no microcontroller).
- To generate an accurate 1 Hz timebase using the NE555 timer in astable mode.
- To implement cascaded decade counting using the CD4026 IC, which combines a Johnson decade counter with a built-in seven-segment decoder/driver.
- To directly drive six common-cathode seven-segment displays without any external BCD-to-segment decoder IC.

- To provide a manual reset facility so the display can be cleared to 000000 at any time.
- To validate the design practically on a breadboard and document the complete build process, stage by stage.

3. Scope of the Project

The clock is an educational, skill-building project for understanding sequential digital logic, decade counting, cascade (carry) propagation, and integrated display driving — concepts that are foundational to digital electronics. Because the CD4026 combines the counter and the display driver in a single package, this build is a good introduction to how much logic can be integrated on-chip, and it can be directly compared against a design built from a separate counter IC and decoder IC (such as the 74LS90 + 74LS47 pair) to see the trade-off between IC count and design flexibility.

3.1 Key Concepts

- Astable multivibrator (NE555): An oscillator mode of the 555 timer with no stable output state — it continuously switches high and low at a rate set by an external resistor-capacitor (RC) network, forming the master clock pulse for the whole circuit.
- Johnson decade counting (CD4026): The CD4026 counts from 0 to 9 on each valid clock edge and internally decodes that count into the correct seven-segment pattern, so no separate BCD-to-seven-segment decoder IC is required.
- Cascading via Carry/Display-Enable Output: Each CD4026 produces a single output pulse once per full 0–9 cycle (its carry/cascade output). Feeding this pulse into the clock input of the next CD4026 makes that stage advance by one only when the previous stage rolls over — the same divide-and-cascade principle used in any multi-digit counter chain.
- Active-high segment drive for common-cathode displays: The CD4026 outputs are active-high, which matches a common-cathode seven-segment display directly (each display's common pin is tied to ground, and a segment lights when its CD4026 output goes high through a current-limiting resistor).
- Debounced vs. undebounced reset: The reset push-button is fed straight into the RESET pin of every CD4026. Like any mechanical switch, its contacts can bounce briefly, which can occasionally cause a slightly irregular reset transition — a normal, well-understood characteristic of unfiltered push-button inputs.

4. System Overview

The circuit is organized into three functional stages that repeat across the six display digits.

4.1 Timebase Generation Stage

An NE555 timer is wired as an astable multivibrator using a resistor–capacitor timing network. This stage produces a continuous square wave that acts as the master clock pulse for the first CD4026 stage. The RC values are chosen and, where a trimmer is used, fine-tuned so the output settles as close as possible to a 1 Hz rate.

4.2 Counting and Display Stage

Six CD4026 ICs are cascaded so that each stage counts 0–9 and, on its tenth pulse, sends a single carry pulse to clock the next stage — the same ripple-cascade principle used to build any multi-digit decimal counter. Each CD4026's seven segment outputs connect, through individual 330 Ω current-limiting resistors, directly to its own common-cathode seven-segment display, so every digit is driven with no intermediate decoder IC.

4.3 Reset Stage

A single push-button switch (S1) is wired to the RESET pin of all six CD4026 ICs in parallel (through the shared reset net). Pressing it drives every counter to zero simultaneously, clearing the full six-digit display back to 000000 without needing to wait for a natural rollover.

5. Components Used

The table below lists the components used in the circuit.

Component	Designator(s)	Qty	Function
CD4026 Decade Counter / 7-Segment Driver IC	IC1–IC6	6	Counts input clock pulses (0–9) and directly drives one common-cathode seven-segment display digit; carry-out cascades to the next stage
Seven-Segment Display (Common Cathode)	DIS1–DIS6	6	Visual digit display, one per CD4026 stage
NE555 Timer IC	IC7	1	Astable multivibrator — generates the 1 Hz master clock pulse
Resistor (330 Ω)	R1–R42 (7 per display)	42	Current-limiting resistor for each display segment
Push-Button Switch	S1	1	Manual reset — clears all counters back to 000000
Regulated Power Supply	VCC / GND	1	5 V DC regulated supply for all ICs and displays
Breadboard(s)	—	As required	Prototyping platform for the full circuit

6. CD4026 and NE555 — Key Pin Functions

6.1 CD4026 Decade Counter / Display Driver

Pin / Signal	Function
CLK (Pin 1)	Clock input — counter advances on each valid clock transition
CLK INH (Pin 2)	Clock inhibit — held low to enable normal counting
RESET (Pin 15)	Active-high reset — clears the internal counter to zero

Pin / Signal	Function
a–g (Pins 3–7, 9, 10)	Seven segment outputs, wired directly to the display through current-limiting resistors
DIS OUT (Pin 11)	Cascade output — toggles once per full 0–9 count, used to clock the next CD4026 stage
CO / Carry Out (Pin 5, shared)	Ripple-carry pulse to the following decade stage
VDD / VSS (Pins 16 / 8)	Supply and ground

6.2 NE555 Timer

- TRIG (Trigger) and THR (Threshold): Sense the capacitor voltage and set the timing of the high/low output transitions.
- DIS (Discharge): Provides the path that discharges the timing capacitor once each cycle.
- OUT (Output, Pin 3): The 1 Hz square wave that drives the first CD4026's clock input.
- CONT (Control Voltage) and RESET: Normally decoupled/tied high respectively for standard astable operation.
- VCC / GND: 5 V supply and ground.

7. Working Principle

7.1 Normal Counting Mode

The NE555 continuously outputs a 1 Hz square wave. Each rising edge clocks the first CD4026 (units digit), which counts 0–9. On its tenth pulse it produces a single cascade pulse that clocks the second CD4026 (tens digit) forward by one, and this pattern repeats through all six stages — the same carry-propagation principle used in any ripple counter chain, just implemented with a single IC type instead of a counter-plus-decoder pair.

7.2 Display Update

Because the CD4026 decodes its own count internally, its seven segment outputs are already in the correct pattern for the current digit at all times — there is no separate decoding step and no additional latching required for the display to stay current.

7.3 Manual Reset

Pressing S1 pulls the RESET line high on every CD4026 simultaneously, clearing all six digits to zero. Releasing the button returns the circuit to normal counting from 000000.

8. Circuit Diagram / Breadboard Photographs

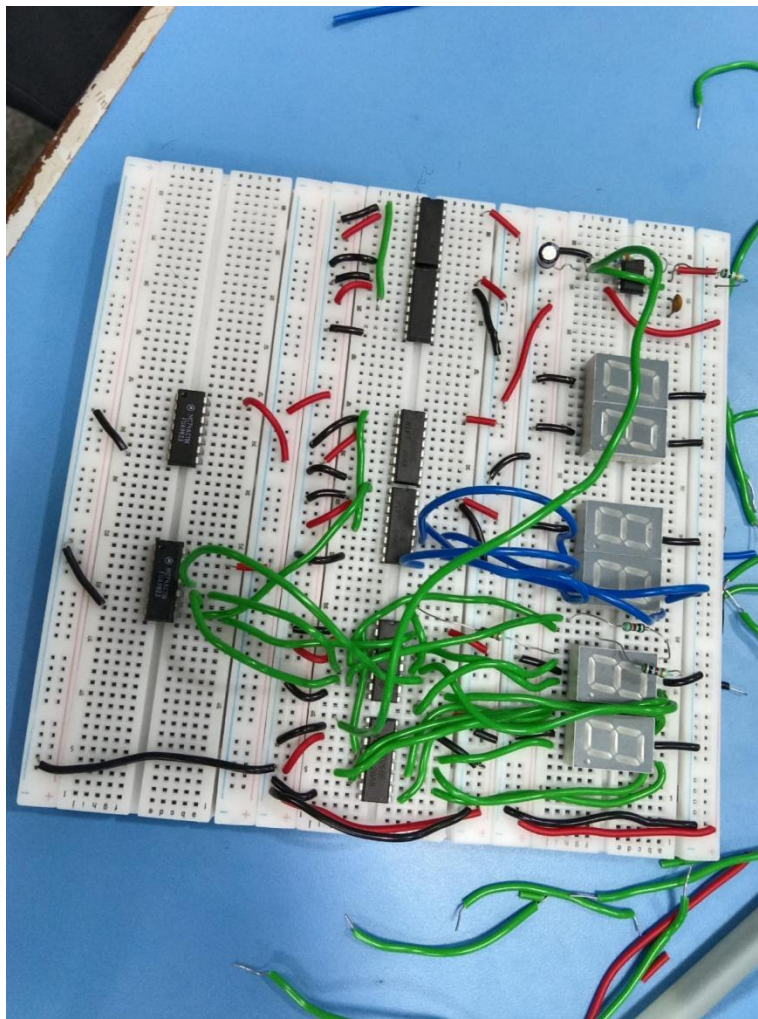


Fig. 1 — Breadboard implementation of the cascaded counter and display stages.

[Insert any additional circuit schematic and breadboard photographs here — for example, the NE555 timebase stage in isolation, a single CD4026 + seven-segment display pair under test, and the completed circuit running.]

9. Testing and Observations

- Stage 1 — Timebase: The NE555 astable circuit was built and probed independently, confirming a stable, repeating 1 Hz pulse before connecting it to any counter.
- Stage 2 — Single digit: One CD4026 + seven-segment display pair was tested first, manually clocked, to confirm correct 0–9 counting and correct segment display.
- Stage 3 — Cascading: Pairs of CD4026 stages were connected and checked for correct carry/rollover behaviour between adjacent digits.
- Stage 4 — Reset logic: S1 was tested to confirm it reliably clears all six digits to zero.
- Stage 5 — Full integration: All six digit stages were connected together with the timebase circuit to form the complete running six-digit display.

10. Applications

- Educational demonstration of sequential logic, decade counting, and integrated display driving without a microcontroller.
- Base platform for a digital stopwatch, event counter, or lap-timer.
- Foundation that can be extended toward a full HH:MM:SS real-time clock by adding reset-at-60 / reset-at-24 logic between digit pairs.

11. Limitations

- As built, the six CD4026 stages form a continuous six-digit up-counter rather than a true HH:MM:SS clock; true time-of-day rollover ($\div 60$ for seconds/minutes, $\div 24$ or $\div 12$ for hours) would need additional reset logic between digit pairs.
- Timekeeping accuracy is limited by the tolerance and thermal drift of the NE555's RC timing network compared to a crystal-oscillator-based design.
- The reset push-button is not debounced, which can occasionally cause a slightly irregular reset transition.
- There is no battery backup, so the count resets to 000000 on power loss.

12. Possible Extensions

- HH:MM:SS rollover logic: Adding a reset-decode gate between each digit pair (reset at '60' for seconds/minutes and at '24' or '12' for hours) would convert this six-digit counter into a true time-of-day clock.
- Crystal-referenced timebase: Replacing the RC-based 555 oscillator with a 32.768 kHz crystal and a binary divider chain would remove thermal-drift error.
- Debounced reset: An RC low-pass filter or an SR-latch debounce circuit on S1 would make the manual reset perfectly clean.
- Battery backup: A small backup supply for the counter ICs alone would let the count survive brief power interruptions.

13. Conclusion

This project demonstrates a complete, self-contained digital counting/clock display built entirely from CMOS logic ICs. It brings together timebase generation (NE555) and integrated decade counting with display driving (CD4026) into one working real-time system, with a manual reset for user control. Building and testing the circuit stage by stage on a breadboard — from the timebase oscillator through to the fully cascaded six-digit display — provided practical, hands-on experience with core digital design concepts for the Chronobit '26 hackathon.

14. References

- Texas Instruments / ON Semiconductor, CD4026B Decade Counter/Divider with Decoded 7-Segment Display Outputs — Datasheet.
- Texas Instruments, NE555 Precision Timer — Datasheet.